

FLUID MECHANICS I.
PROBLEM 2 (10 POINTS).

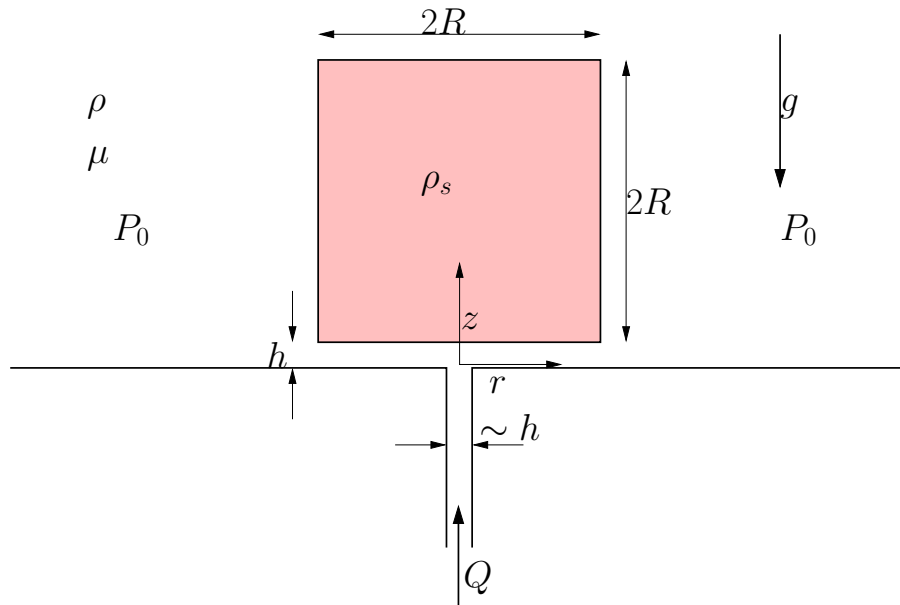
16-01-2019
Time: 75 minutes

NAME:.....

GRADE:.....

The cylinder of radius R and height $2R$ shown in the figure is submerged in an infinite tank of liquid of density ρ and viscosity μ . The cylinder, of density $\rho_s > \rho$, is sustained at a constant height $h \ll R$ by the effect of a narrow jet of volumetric flow rate Q and radius of the order of h , injected at the center of the base of the cylinder (see the sketch). Assuming the fluid movement is negligible in the bulk of the tank ($P_0 = \text{const.}$) and the flow below the cylinder is dominated by viscosity, we want to determine the dependence of h as a function of Q . To this end, the following sequence is suggested:

1. Give the criterion, as a function of the known parameters, for the flow in the gap of width h between the cylinder and the bottom of the tank to be dominated by viscosity.
2. Determine the velocity profile as a function of the local reduced pressure gradient $\partial P / \partial r$.
3. Obtain the local flow rate $q(r)$ at a generic radial position r
4. Use the continuity equation to derive an equation for the reduced pressure P , with all needed boundary conditions. Obtain P .
5. Determine h



① ~~$\rho \frac{\partial v_r}{\partial t} + \rho v_r \frac{\partial v_r}{\partial r} + \rho v_z \frac{\partial v_r}{\partial z} = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_r) \right) + \frac{\partial^2 v_r}{\partial z^2} \right]$~~

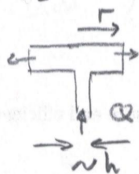
STEADY $\frac{\rho v_c^2}{R} \quad \frac{\rho v_c v_r}{h} \quad \frac{\Delta r R}{R} \quad \frac{\mu v_c}{R^2} \ll \frac{\mu v_c}{h^2}$

$$\frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{\partial^2 v_r}{\partial z^2} = 0$$

$$\frac{v_c}{R} \sim \frac{v_c}{h} \rightarrow v_c \sim v_r \frac{h}{R} \ll v_r$$

$$\frac{\text{CONVECTION}}{\text{VISCOUS}} \sim \frac{\rho v_c^2 \frac{h^2}{R}}{R \mu v_c} = \frac{\rho v_c h}{\mu} \frac{h}{R} \ll 1$$

WHERE v_c CAN BE ESTIMATED FROM CONTINUITY:



$$2\pi r h v_r \sim Q \rightarrow v_r \sim \frac{Q}{r h}$$

$$\text{FOR } r \sim h \quad v_r \sim \frac{Q}{h^2} \Rightarrow \text{CRITERION } \frac{\rho Q}{\mu R} \ll 1$$

②
$$\left. \begin{aligned} 0 &= -\frac{\partial p}{\partial r} + \mu \frac{\partial^2 v_r}{\partial z^2} \\ z=0: v_r &= 0 \\ z=h: v_r &= 0 \end{aligned} \right\} \quad \left. \begin{aligned} \frac{\partial v_r}{\partial z} &= \frac{1}{\mu} \frac{\partial p}{\partial r} z + A \\ v_r &= \frac{1}{2\mu} \frac{\partial p}{\partial r} z^2 + A z + B \\ 0 &= \frac{1}{2\mu} \frac{\partial p}{\partial r} h^2 + A h \end{aligned} \right\} \quad \left\{ \begin{aligned} v_r &= -\frac{1}{2\mu} \frac{\partial p}{\partial r} (h^2 - z^2) \end{aligned} \right.$$

③
$$Q = \int_0^h 2\pi r v_r dz = -\frac{2\pi r}{2\mu} \frac{\partial p}{\partial r} \left[h \frac{z^2}{2} - \frac{z^3}{3} \right]_0^h = -\frac{\pi h^3 r}{6\mu} \frac{\partial p}{\partial r}$$

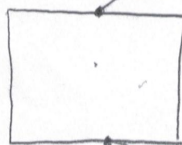
④ $q(r) = Q = \text{const.} \rightarrow \left\{ \begin{aligned} Q &= -\frac{\pi h^3 r}{6\mu} \frac{\partial p}{\partial r} \\ r=R: p &= p_0 \end{aligned} \right. \rightarrow \frac{\partial p}{\partial r} = -\frac{6\mu Q}{\pi h^3 r}$

$$p = -\frac{6\mu Q}{\pi h^3} \ln r + C$$

$$p_0 = -\frac{6\mu Q}{\pi h^3} \ln R + C$$

$$p = p_0 - \frac{6\mu Q}{\pi h^3} \ln \frac{r}{R}$$

⑤ $p_r = p_0 - s_g \chi(r, R)$



$$p_r = p - s_g h = p_0 - s_g h - \frac{6\mu Q}{\pi h^3} \ln \frac{r}{R}$$

$$F_{L \rightarrow S} = \int_0^R 2\pi r (p_r - p_r) \cdot dr$$

$$W = 2\pi R^2 s_g g$$

h is determined by the balance

$$W = F_{L \rightarrow S}$$

$$F_{L \rightarrow S} = +2\pi R^3 S g - 2\pi \int_0^R \frac{6\mu Q}{\pi h^3} r \ln \frac{r}{R} dr =$$

$$= +2\pi R^3 S g - \frac{12\mu Q R^2}{h^3} \underbrace{\int_0^1 s \ln s ds}_{\left[\frac{s^2}{2} \ln s - \frac{s^2}{4} \right]_0^1 = -\frac{1}{4}} = 2\pi S g R^3 + \frac{3\mu Q R^2}{h^3}$$

$$F_{L \rightarrow S} = W \Rightarrow \frac{3\mu Q R^2}{h^3} = 2\pi R^3 g (S_s - S) \rightarrow h = \left[\frac{3\mu Q R^2}{2\pi R^3 g (S_s - S)} \right]^{1/3}$$